Exoplanet Transit Depth: Theory and Distribution

1. Abstract/Introduction

In this project, I explore the exoplanet transit depth using data from the NASA Exoplanet Archive. A transit happens when a planet passes in front of its host star and blocks part of the star's light. Therefore, the observed brightness of the star decreases for a short time. This decrease in brightness is called transit depth.

A simple model predicts that the transit depth should follow the equation below:

$$\delta(\%) \approx 100 \left(\frac{R_p}{R_*}\right)^2$$

where R_p is the planet radius and R_* is the stellar radius.

My goal was to study exoplanet transit depth from two perspectives. First, I tested whether the observed transit depth values in the archive follow the relation above. Secondly, I studied how exoplanets are distributed across different transit depth ranges.

2. Tools/Resources

The main Python libraries I used were NumPy, Pandas, Matplotlib, and SciPy.

I used data from the NASA Exoplanet Archive. I chose the Exoplanet which was found in 2025 instead of 2026 because NASA hasn't found quite a lot of new Exoplanets yet (Unfortunately, NASA reported about 110 new exoplanets on April 23, 2026, but by that time I had already completed my analysis and report).

The main columns I used were pl_trandep, which gives the observed transit depth in percent, pl_ratror, which gives the ratio of planet radius to stellar radius, and pl_trandeperr1, which gives the upper uncertainty in transit depth. I also used default_flag to keep only NASA's default parameter set for each planet.

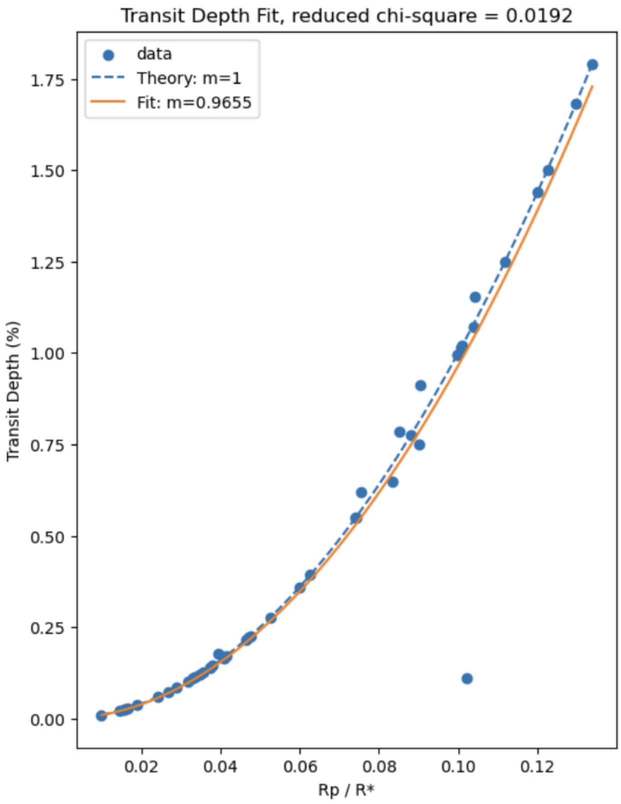
I filtered the dataset so that I only kept rows with default_flag = 1 and removed rows with missing values in the columns needed for the analysis. This allowed me to work with one main record for each planet instead of including multiple published values for the same object.

3. Testing the Transit Depth Formula

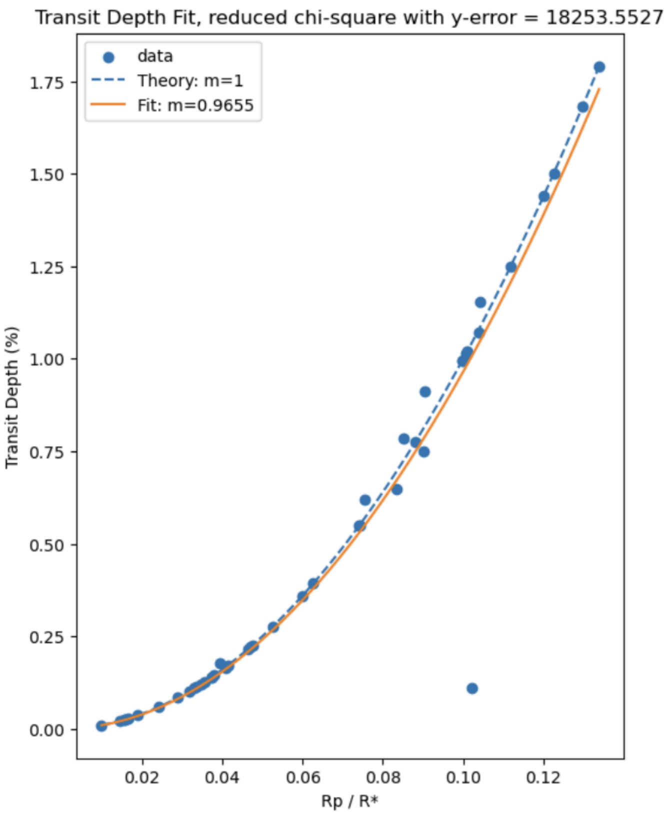
In order to test the relation, I fitted the model:

$$\delta(\%) = m \cdot 100\left(\frac{R_p}{R_*}\right)^2$$

where m is my fitting parameter.



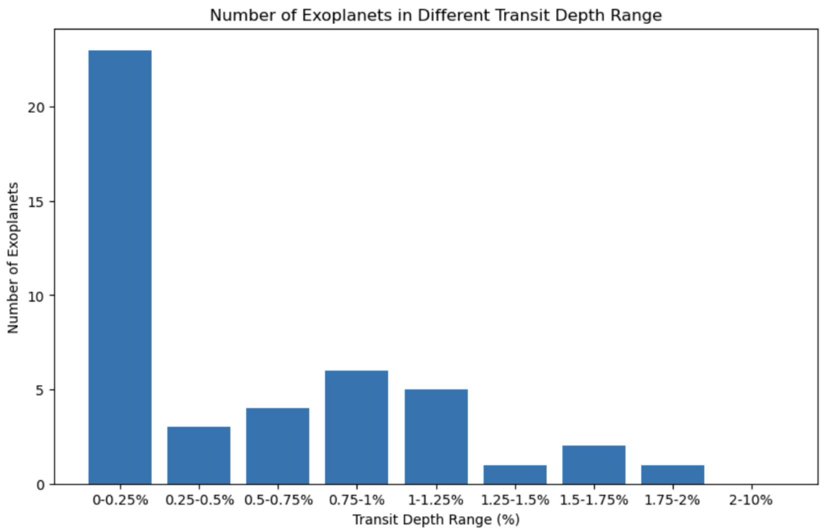
I used curve_fit from SciPy to find the best-fit value of m. As shown in the graph above, the m value I got is 0.9655 which is very close to 1, but my reduced chi-square is very small compared to 1. After examining my code, I found that I first calculated a simplified reduced chi-square value using the same method from Homework 9, where it assumes the uncertainty is 1 for each data point. According to my current reduced chi-square 0.0192, the real uncertainty might be smaller than 1 to allow the reduced chi-square closer to 1. As a result, I calculated a more standard reduced chi-square by including the uncertainty in the y-direction from pl_trandeperr1.



I redefined my data and performed the fit again here because when I calculated the reduced chi-square using the original data set, I gained an undefined value at first. Then, I noticed that there are some data points that don't have the pl_trandeperr1 value. After removing those points, I got the same fitting parameter m = 0.9655 as I did from the previous fit, meaning that deleting those undefined values is rare and doesn't affect the fitting. This m shows that the observed transit depth data follow the trend very well. In other words, the data support that the transit depth increases roughly like $(\frac{R_p}{R_*})^2$. However, compared to 1, the reduced chi-square value is very large !!! This might suggest that the real data do not match this simple model closely enough compared with the very small published uncertainties.

4. Distribution of Transit Depth

In the second part of the project, I studied how exoplanets are distributed across different transit depth ranges.



The bar chart above shows that most exoplanets in my dataset have relatively small transit depths. The number of planets was highest in the lowest transit-depth ranges and generally decreased as transit depth became larger. This result is very interesting to me because initially I thought that larger transit depth makes a transit easier to detect in a simple sense. It is like blocking a light bulb with a book instead of a small piece of paper. The larger object blocks more light, so it is much easier to notice that something is there. However, the archive still contains many more planets with small transit depths than planets with large ones in 2025. I guess that even though higher transit depth might be easier to find, a higher transit depth also means to have a system that has a large planet and a small host star. Maybe this kind of combination is less common in the whole universe.

5. Conclusion

Overall, I enjoyed this project because it let me connect a simple equation to real exoplanet data. One challenge was understanding the reduced chi-square results and handling missing uncertainty values, but that helped me better understand the fitting process. I also found that the model follows the trend well with m = 0.9655, and that most exoplanets in my dataset have relatively small transit depths.